C0MPILED_10 · DC HACKATHON 2026

NASA EXOPLANET ARCHIVE

Exoplanet Query Interface

The universe has answers. Ask them.

A natural language AI interface over real NASA exoplanet data — built on modern PostgreSQL to replace aging legacy systems.

THE PROBLEM

Legacy Infrastructure Is Holding NASA Back

01 — SILOED DATA

Aging SQL Server Systems

NASA's scientific data lives in legacy Microsoft SQL Server infrastructure — inaccessible to modern AI tooling and downstream applications.

02 — EXPERT-ONLY ACCESS

Insights Go Undiscovered

Researchers and the public can't query the data without deep SQL expertise. Valuable discoveries remain locked behind a technical barrier.

03 — BLOCKED AI PIPELINE

No Path to Modern Tooling

Legacy infrastructure prevents machine learning, open data APIs, real-time dashboards, and LLM integrations from ever being built.

04 — STALLED MIGRATION

No Compelling Proof of Concept

Modernization efforts have stalled for years — lacking a working demonstration to justify the investment and align stakeholders.

THE SOLUTION

Modernize. Query. Discover.

NASA Archive
Exoplanet data

→

PostgreSQL
Structured & indexed

→

LLM via Groq
Text-to-SQL

→

Plain English
For everyone

Legacy Migration

Automated pipeline pulls NASA Exoplanet Archive data into PostgreSQL — structured, indexed, and ready for AI queries from day one.

Natural Language Query

Ask anything in plain English. The LLM writes SQL, executes it, and explains results in a vivid 2–3 sentence answer — no expertise needed.

REAL NASA DATA · LIVE DATABASE

20,000+ Records. 5 Tables. One Question.

6,273
Confirmed Exoplanets

7,927
TESS Candidates

4,701
Host Stars

1,473
Asteroid Events

EXAMPLE QUERIES

"Which mission found the most planets?
→ Kepler, with 2,784 confirmed discoveries.

"Find Earth-sized planets in the habitable zone.
→ Results with orbital period, temperature & distance.

"Which asteroid passed closest to Earth in 2024?
→ Name, miss distance, and velocity in seconds.

"What is the most common exoplanet type?
→ Super-Earths, comprising over 30% of all confirmed planets.

THE VISION

This is more than a demo. It's proof that decades of siloed government data can be unlocked for researchers, policymakers, and the public — through AI.

Immediate Impact

Researchers and citizens can query 6,273 worlds instantly — no SQL, no middleman, no waiting for a data team.

Scales to Any Agency

The same pipeline applies to IRS tax data, EPA environmental records, or any legacy SQL Server system in government.

PostgreSQL Migration | Text-to-SQL | Open Government Data | AI Infrastructure | NASA · IRS · EPA